MA3923 Monte Carlo Simulation – Lab 10

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1 Question 1: Fitting a GBM to stock data

1.1 Approach

Let $P_0, \dots, P_n$ be the daily adjusted closing prices. The daily returns are $R_i = P_i/P_{i-1} - 1$ for $i = 1, \dots, n$. Calling their sample mean A and sample standard deviation B, the annualised parameters are

$$\mu = 252A, \qquad \sigma = \sqrt{252}\,B, \qquad T = \frac{n}{252}, \qquad S(0) = P_0.$$

I then treat the price as a GBM, $dS = \mu S\,dt + \sigma S\,dW$, which has the exact solution

$$S(T) = S(0)\exp\Big\{\big(\mu - \tfrac{1}{2}\sigma^2\big)T + \sigma\sqrt{T}\,Z\Big\}, \qquad Z \sim N(0,1),$$

so $\mathbb{E}[S(T)] = S(0)e^{\mu T}$. Following Lab 09, I simulate $S(T)$ a large number of times, estimate $\mathbb{E}[S(T)]$ by the sample mean with a 95% confidence interval, and compare it with the actual last price P_n. I also simulate a few full daily paths and plot them against the real price history.

1.2 Results

I used the stock `TICKER` with adjusted closing prices from START to END, giving $n =$ ___ daily returns and $T = n/252 =$ ___ years. The values below come from running `q1.py`.

Quantity	Value
$S(0)$	---
Mean daily return A	---
Std of daily returns B	---
$\mu = 252A$	---
$\sigma = \sqrt{252}B$	---
Estimated $\mathbb{E}[S(T)]$ (100000 simulations)	---
95% margin of error	---
Actual $S(T)$	---
Relative error (%)	---

1.3 Observations

The simulated mean agrees with the exact value $S(0)e^{\mu T}$, so the sampler is doing what it should. That said, the real price history is just one realisation of the model, so there is no reason for the

actual final price to land on the model mean. Over a long horizon $S(T)$ is lognormal and heavily right-skewed, which makes the mean sit well above the median. For that reason the percentile of the actual price in the simulated distribution is a fairer comparison than the mean alone.

Two other points are worth keeping in mind. First, μ is estimated from the average return, and its standard error is roughly $\sigma/\sqrt{T}$, which is large. Because the forecast grows like $e^{\mu T}$, a small error in μ turns into a big error in the predicted price when T is long. Second, GBM assumes constant μ and σ and normally distributed log-returns. Real stocks show volatility clustering, sudden crashes and fatter tails, so the fitted model tends to understate how rough the actual path can get.

2 Question 2: Jump diffusion

2.1 Setup

The process is simulated with

$$X(t_{i+1}) = X(t_i) + \left(\mu - \tfrac{1}{2}\sigma^2\right)(t_{i+1} - t_i) + \sigma\left[W(t_{i+1}) - W(t_i)\right] + \sum_{j=N(t_i)+1}^{N(t_{i+1})} \log Y_j,$$

on $[0, 1]$ with $X(0) = 5$, $\mu = 0.06$, $\sigma = 0.3$ and 1000 sub-intervals, so $\Delta t = 0.001$. In each step the number of jumps is Poisson with mean $\lambda\Delta t$, and $Y_j = 0.1Z_j + 1$ with $Z_j \sim N(0, 1)$, which makes $\log Y_j$ roughly $N(-0.005,\, 0.1^2)$. I used $\lambda \in \{0.01, 0.05, 0.1, 0.5, 1\}$.

For part (a) I generated the path with the Wiener part only. For part (b) I added jumps for each λ, reusing the same Brownian increments as in part (a), so any gap between a jump path and the no-jump path comes purely from the jumps. Part (c) puts all six paths in one plot, and for part (d) I repeated everything with four more independent sets, giving five sets in total (Figures 1–5).

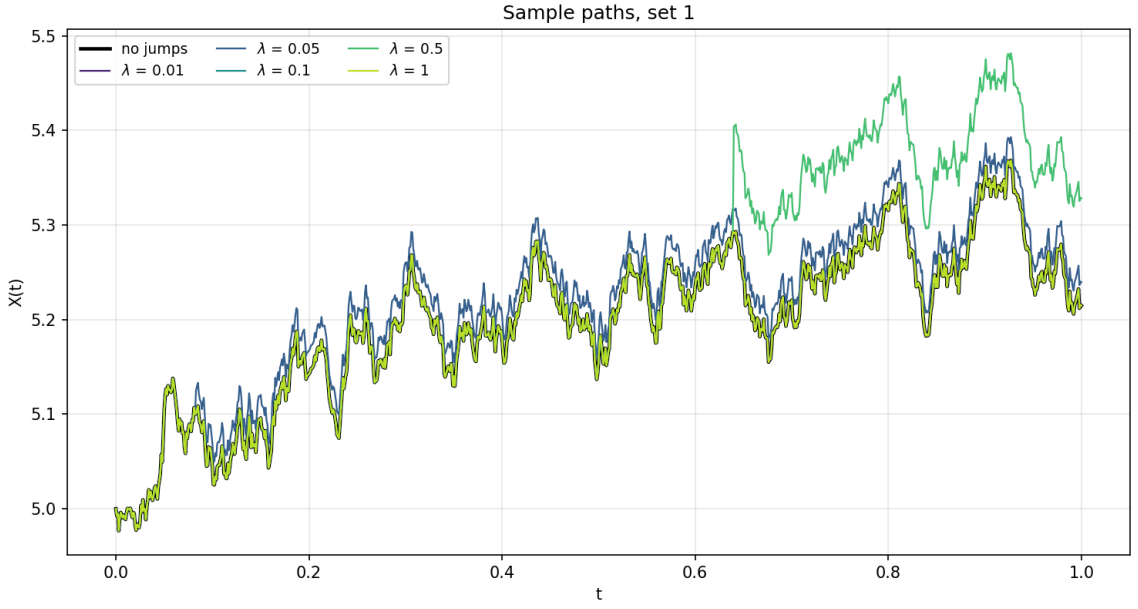


Figure 1: Set 1: the no-jump path (black) and the jump paths for each λ.

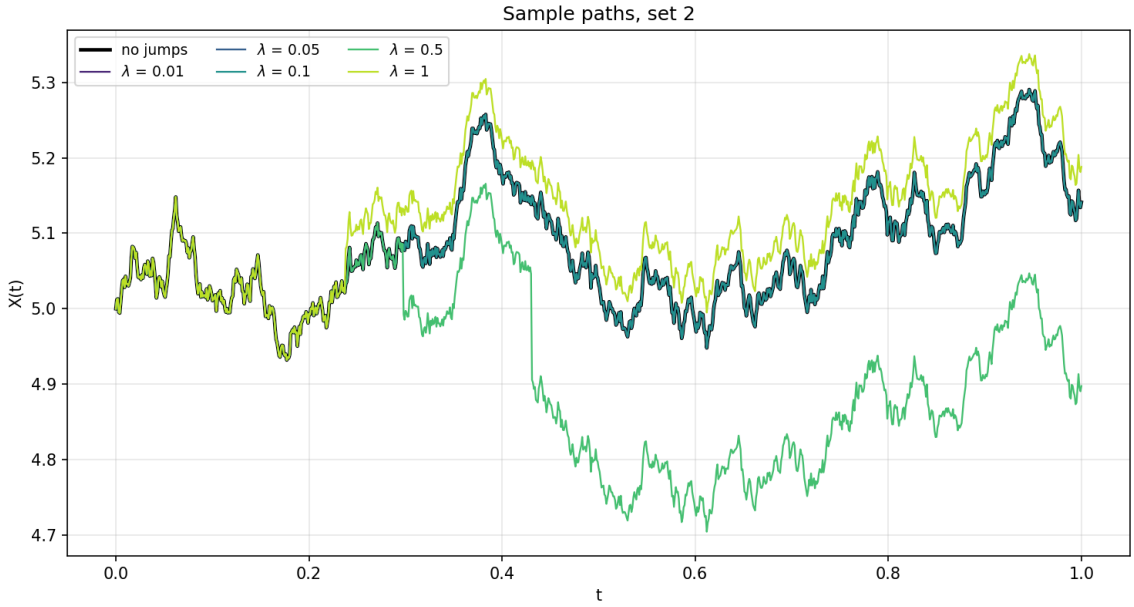


Figure 2: Set 2.

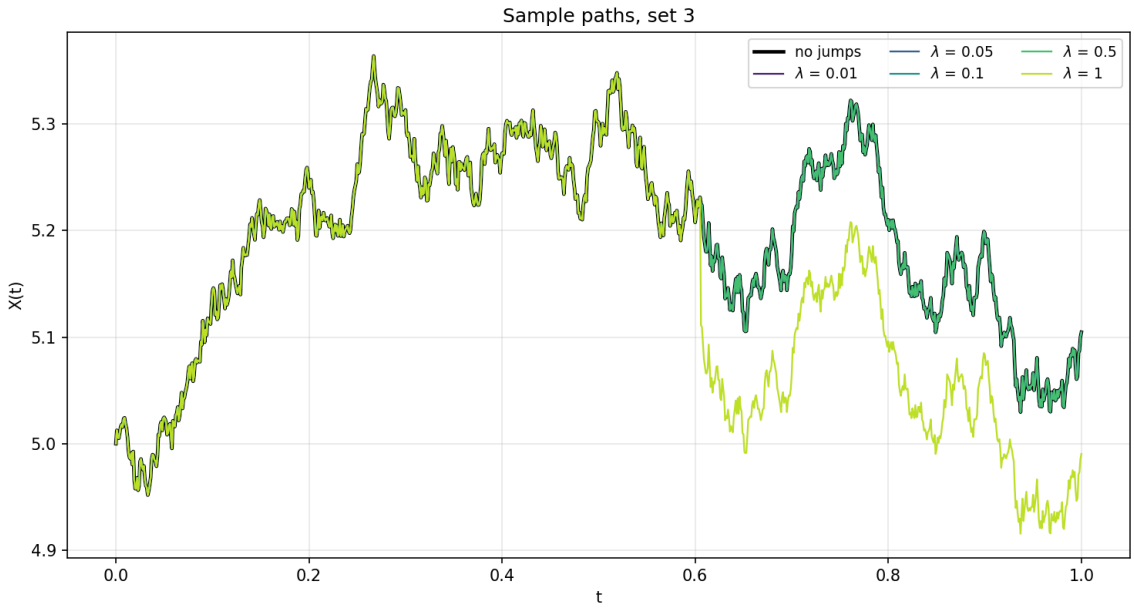


Figure 3: Set 3.

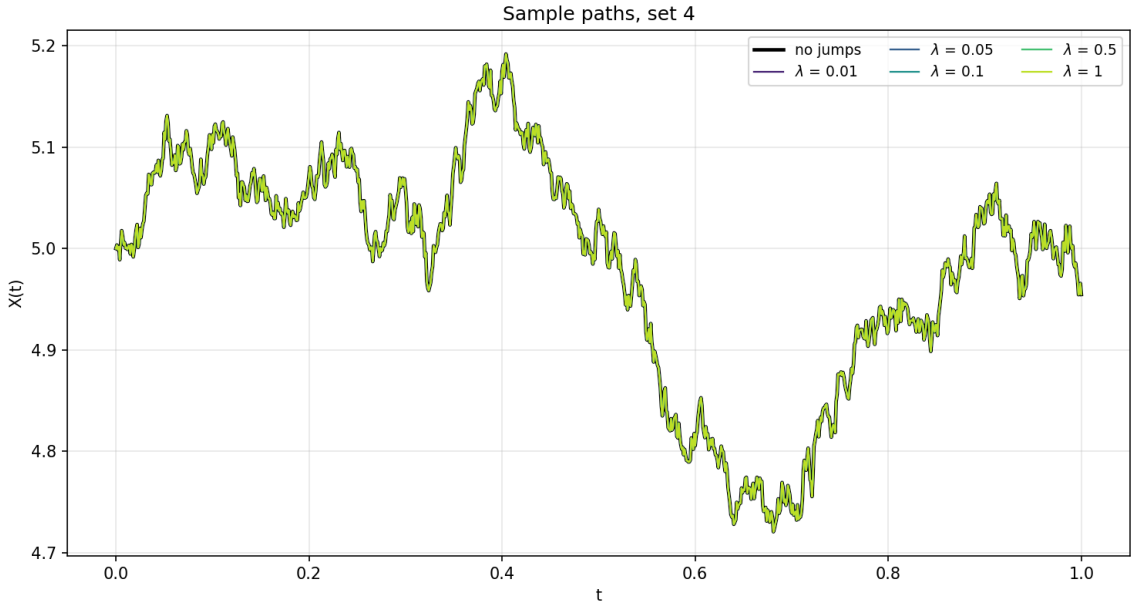


Figure 4: Set 4.

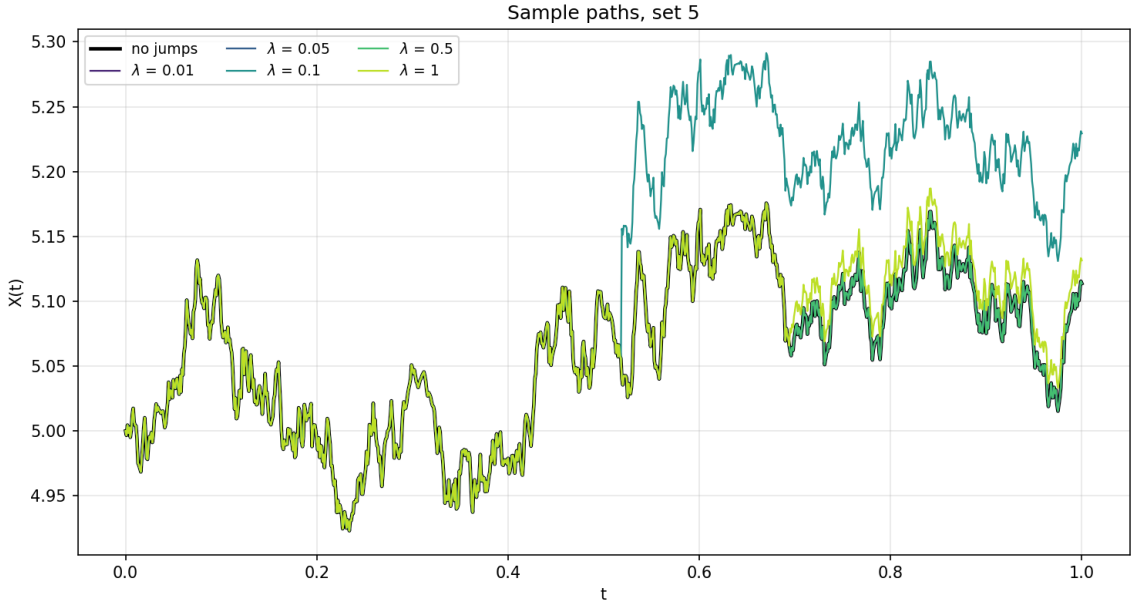


Figure 5: Set 5.

2.2 Numbers

Table 1 gives, for every path, the number of jumps, the final value, and the largest gap from the no-jump path driven by the same noise.

Set	λ	Jumps	$X(1)$ with jumps	$X(1)$ no jumps	$\max_t \lvert X - X_{\text{nj}} \rvert$
1	0.01	0	5.2149	5.2149	0.0000
1	0.05	1	5.2395	5.2149	0.0246
1	0.1	0	5.2149	5.2149	0.0000
1	0.5	1	5.3282	5.2149	0.1134
1	1	0	5.2149	5.2149	0.0000
2	0.01	0	5.1411	5.1411	0.0000
2	0.05	0	5.1411	5.1411	0.0000
2	0.1	0	5.1411	5.1411	0.0000
2	0.5	2	4.8975	5.1411	0.2435
2	1	1	5.1880	5.1411	0.0470
3	0.01	0	5.1046	5.1046	0.0000
3	0.05	0	5.1046	5.1046	0.0000
3	0.1	0	5.1046	5.1046	0.0000
3	0.5	0	5.1046	5.1046	0.0000
3	1	1	4.9904	5.1046	0.1142
4	0.01	0	4.9547	4.9547	0.0000
4	0.05	0	4.9547	4.9547	0.0000
4	0.1	0	4.9547	4.9547	0.0000
4	0.5	0	4.9547	4.9547	0.0000
4	1	0	4.9547	4.9547	0.0000
5	0.01	0	5.1137	5.1137	0.0000
5	0.05	0	5.1137	5.1137	0.0000
5	0.1	1	5.2294	5.1137	0.1157
5	0.5	0	5.1137	5.1137	0.0000
5	1	1	5.1316	5.1137	0.0178

Table 1: Summary of the paths in the five sets.

To see the average effect of the jumps I also ran 20000 independent replications for each λ, and for $\lambda = 0$, and recorded the mean and standard deviation of $X(1)$.

λ	mean of $X(1)$	std of $X(1)$
0	5.0164	0.3004
0.01	5.0149	0.3003
0.05	5.0178	0.3001
0.1	5.0160	0.3019
0.5	5.0136	0.3073
1	5.0059	0.3197

Table 2: Mean and standard deviation of $X(1)$ over 20000 replications.

2.3 Observations

The jumps are easy to spot on the plots. The diffusion moves about $\sigma\sqrt{\Delta t} \approx 0.0095$ per step, while a typical jump is around 0.1, roughly ten times bigger, so a jump appears as a sharp vertical step in an otherwise wiggly path. For example, in set 2 the $\lambda = 0.5$ path ends up as much as 0.24 away from the no-jump path.

For small λ very little happens. The expected number of jumps on $[0, 1]$ is λ, and for $\lambda = 0.01$ there were no jumps at all in any of the five sets, so those paths lie exactly on top of the black no-jump path and are hidden under the later curves. Over the five sets the total jump counts for $\lambda = 0.01, 0.05, 0.1, 0.5, 1$ were $0, 1, 1, 3, 3$, compared with expected values of $0.05, 0.25, 0.5, 2.5, 5$.

The paths are also not nested in λ. Each λ gets its own Poisson draws, so a path with $\lambda = 1$ can have no jumps while one with $\lambda = 0.5$ has two. The chance of no jumps is $e^{-\lambda}$, about 0.37 for $\lambda = 1$, which fits the several $\lambda = 1$ paths with zero jumps.

The jumps barely change the mean. Since $\mathbb{E}[\log Y] \approx -0.005$, a jump shifts the level by almost nothing on average, and the estimated means of $X(1)$ stay between 5.006 and 5.018 for all λ. The Monte Carlo standard error is about 0.002, and the diffusion-only value is $5 + \mu - \sigma^2/2 = 5.015$. Jumps are symmetric in sign with a very slight downward tilt, and both directions appear: the $\lambda = 0.5$ jump in set 1 goes up, while the ones in set 2 go down.

The spread is where jumps show up in the statistics. They add variance of roughly $\lambda\,\mathbb{E}[(\log Y)^2] \approx 0.01\lambda$ to the diffusion variance $\sigma^2 = 0.09$, so the standard deviation of $X(1)$ rises from about 0.300 at $\lambda = 0$ to about 0.320 at $\lambda = 1$. This is close to $\sqrt{0.09 + 0.01} \approx 0.316$, with the small gap explained by simulation noise.

Finally, across the five sets the no-jump final value ranges from 4.95 to 5.21, so the Wiener part is still the main source of variation between paths. With this jump size, jumps are a secondary effect. They would matter more with a larger λ or a larger standard deviation for Y_j.